

8 (a) (constant) probability of decayM1
 per unit time A1 [2]
(reference to decay of isotope / mass / sample / nuclide, allow max 1 mark)

(b) *either* when time = $t_{1/2}$, $N = \frac{1}{2}N_0$ M1
or $\frac{1}{2}N_0 = N \exp(-\lambda t_{1/2})$ M1
either $2 = \exp(\lambda t_{1/2})$ M1
or $\frac{1}{2} = \exp(-\lambda t_{1/2})$ M1
 (taking logs), $\ln 2 = 0.693 = \lambda t_{1/2}$ A1 [3]

(c) $A = \lambda N$
 $1.8 \times 10^5 = N \times (0.693 / \{1.66 \times 10^8\})$ C1
 $N = 4.3 \times 10^{13}$
 mass = $60 \times (N / N_A)$ or $60 \times N \times u$ C1
 $= (60 \times 4.3 \times 10^{11}) / (6.02 \times 10^{23})$
 $= 4.3 \times 10^{-9} \text{ g}$ A1 [3]

[Total: 8]

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Section B

0 (a) as reduces gain